14-1b The Revenue of a Competitive Firm

Like most other firms in the economy, a firm in a competitive market tries to maximize profit (total revenue minus total cost). To see how it does this, let's begin by considering the revenue of a competitive firm: the Vaca Family Dairy Farm.

The Vaca Farm produces a quantity of milk, Q, and sells each unit at the market price, P. The farm's total revenue is $P \times Q$. For example, if a gallon of milk sells for $\$6$ and the farm sells $1{,}000$ gallons, its total revenue is $\$6{,}000$.

Because the Vaca Farm is small compared with the world market for milk, it takes the price as given by market conditions. This means that the price of milk does not depend on the number of gallons that the Vaca Farm produces and sells. If the Vacas double the amount of milk they produce to $2{,}000$ gallons, the price of milk remains the same, and their total revenue doubles to $\$12{,}000$. As a result, total revenue is proportional to the amount of output.

Table 1 shows the revenue for the Vaca Family Dairy Farm. Columns (1) and (2) show the amount of output the farm produces and the price at which it sells its output. Column (3) is the farm's total revenue. The table assumes that the price of milk is $\$6$ a gallon, so total revenue is $\$6$ times the number of gallons.

Table 1

Total, Average, and Marginal Revenue for a Competitive Firm

(1)	**(2)**	**(3)**	**(4)**	**(5)**
Quantity (**Q**)	**Price** (**P**)	**Total Revenue** $(\boldsymbol{TR = P \times Q})$	**Average Revenue** $(\boldsymbol{AR = TR/Q})$	**Marginal Revenue** $(\boldsymbol{MR = \Delta TR/\Delta Q})$
1 gallon	$6	$6	$6	
				$6
2	6	12	6	

(1)	(2)	(3)	(4)	(5)
Quantity (***Q***)	**Price** (***P***)	**Total Revenue**	**Average Revenue**	**Marginal Revenue**
				6
3	6	18	6	
				6
4	6	24	6	
				6
5	6	30	6	
				6
6	6	36	6	
				6
7	6	42	6	
				6
8	6	48	6	

Just as the concepts of average and marginal were useful in the preceding chapter when analyzing costs, they are also useful when analyzing revenue. To see what these concepts tell us, consider these two questions:

- How much revenue does the farm receive for the typical gallon of milk?
- How much additional revenue does the farm receive if it increases production of milk by 1 gallon?

Columns (4) and (5) in Table 1 answer these questions.

Column (4) in the table shows **average revenue** (total revenue divided by the quantity sold), which is total revenue [from column (3)] divided by the amount of output [from column (1)]. Average revenue tells us how much revenue a firm receives for the typical unit sold. In Table 1, you can see that average revenue equals $\$6$, the price of a gallon of milk. This illustrates a general lesson that applies not only to competitive firms but to other firms as well. Average revenue is total revenue $(P \times Q)$ divided by the quantity (Q). Therefore, *for all types of firms, average revenue equals the price of the good.*

Column (5) shows **marginal revenue** (the change in total revenue from an additional unit sold), which is the change in total revenue from the sale of each additional unit of output. In Table 1, marginal revenue equals $\$6$, the price of a gallon of milk. This result illustrates a lesson that applies only to firms in competitive markets. Because total revenue is $P \times Q$ and P is fixed for a competitive firm, when Q rises by 1 unit, total revenue rises by P dollars. *Therefore, for competitive firms, marginal revenue equals the price of the good.*

QuickQuiz

1. A perfectly competitive firm
    a. chooses its price to maximize profits.
    b. sets its price to undercut other firms selling similar products.
    c. takes its price as given by market conditions.
    d. picks the price that yields the largest market share.

2. When a perfectly competitive firm increases the quantity it produces and sells by 10 percent, its marginal revenue ____ and its total revenue rises by ____ .
    a. falls; less than 10 percent
    b. falls; exactly 10 percent
    c. stays the same; less than 10 percent
    d. stays the same; exactly 10 percent